

Forwarding Score: A New Metric for Assessing the Quality of Internet Route Surveys

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Introduction

Internet route surveys like **CAIDA Ark** and **IPRS** map the Internet through large-scale traceroute measurements. Traditionally, their quality has been judged by *topology coverage* (how many nodes and links they reveal).

However, this focus on structural discovery overlooks a key dimension: **how well these systems capture routing behavior**.

We introduce the **Forwarding Score**, a new metric designed to quantify how comprehensively a survey measures router forwarding actions.

Forwarding Score Definition

Each element of forwarding information is a three-tuple:

$$\varphi = (a_n, D, a_f)$$

where:

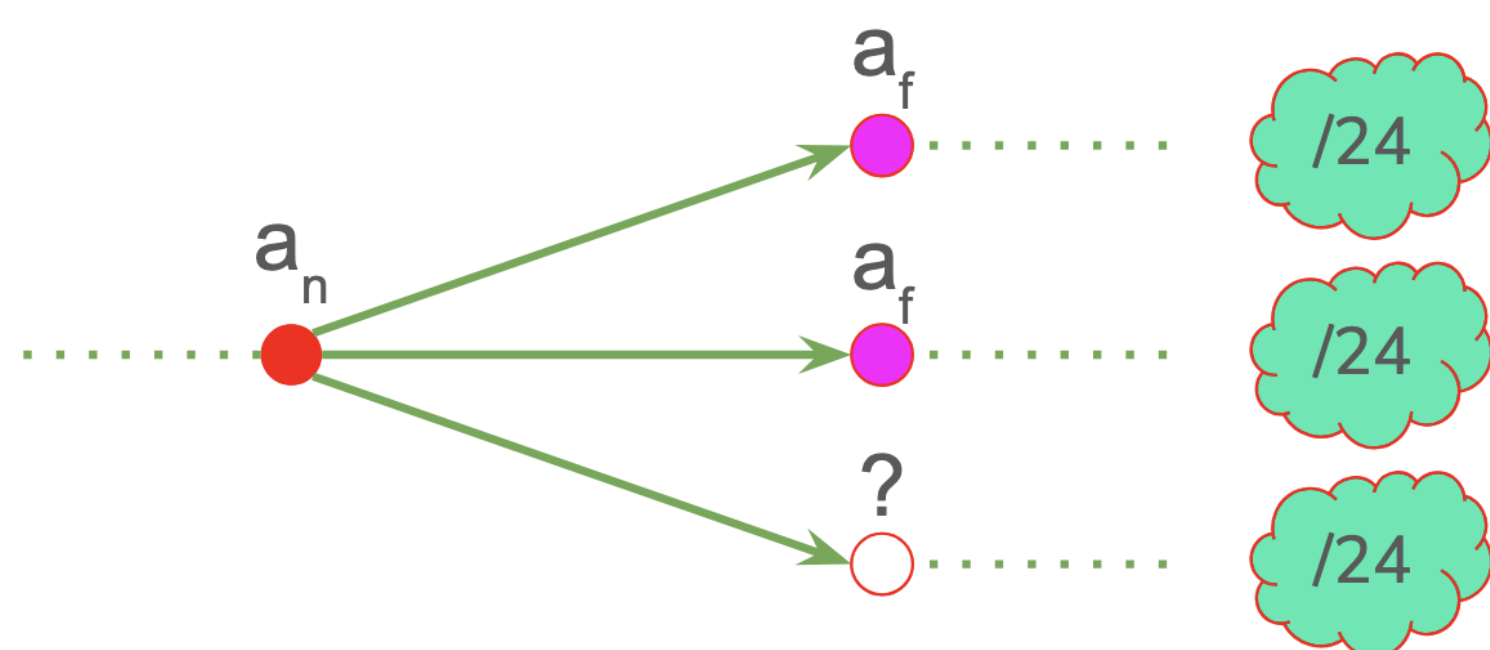
- a_n : near (forwarding) address,
- D : destination prefix (/24 for IPv4),
- a_f : far address.

The **Forwarding Score** of address a counts the number of distinct destination prefixes D for which forwarding information exists:

$$F(a) = |\{D : \exists a_f : (a, D, a_f) \in \Phi\}|$$

It reflects how much of the Internet's forwarding behavior is observed from a .

Example:



Implementation: MPAT Tool

We built **MPAT**, an open-source analysis tool written in Go and backed by **ClickHouse**.

- Downloads traceroute data from different sources.
- Extracts forwarding information elements from traceroute sequences.
- Removes duplicates across vantage points and destinations.
- Computes forwarding scores for suitable addresses.

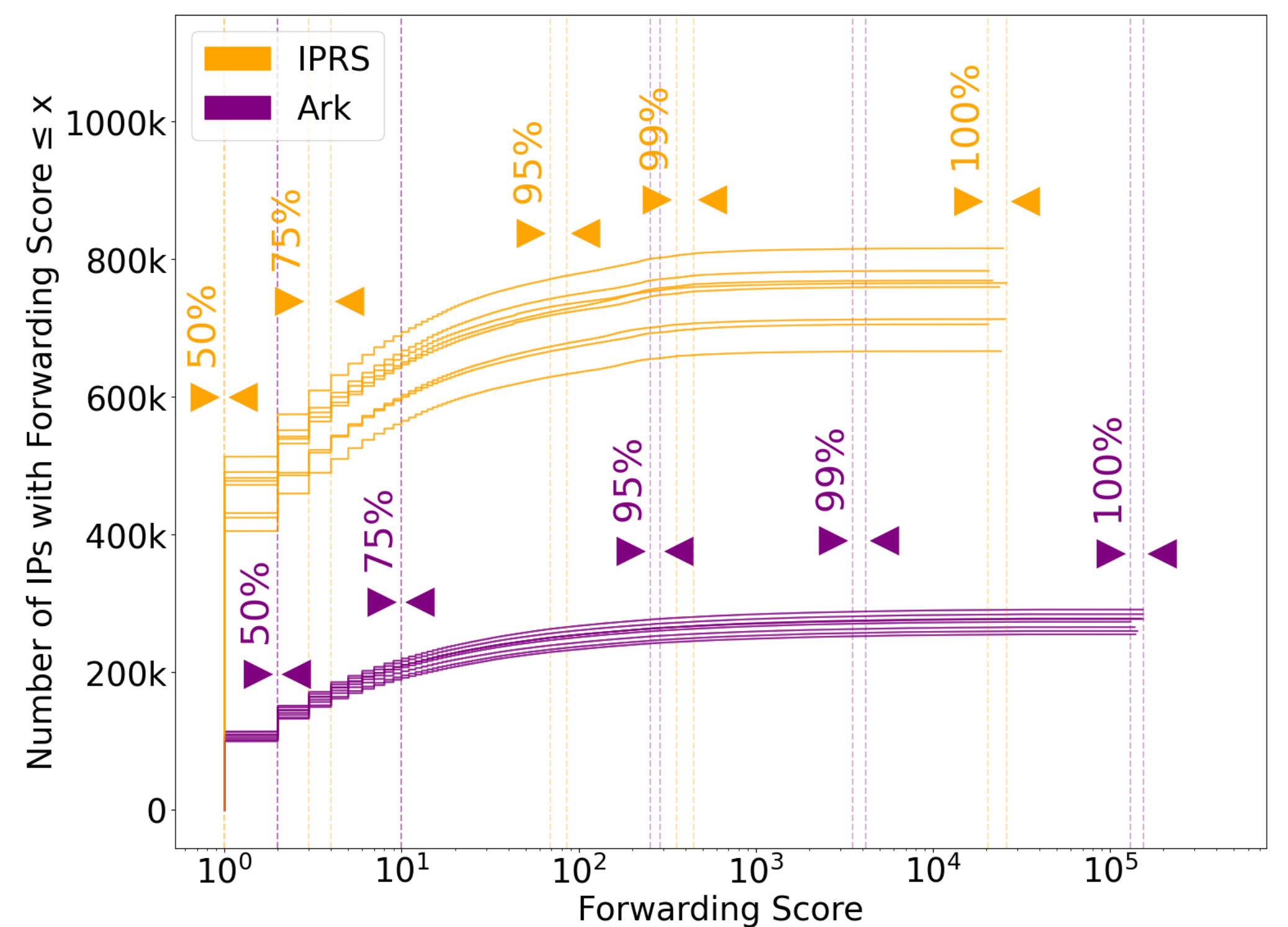
All code and data are publicly available.

Results

We computed forwarding scores for eight days of data (**May 10–17, 2025**) from both Ark and IPRS.

IPRS observed more addresses with nonzero forwarding scores (670k–820k) than **Ark** (260k–290k).

Ark reached higher maximum scores (130k–150k) than IPRS (20k–26k).



Discussion

The **Forwarding Score** reveals trade-offs invisible to topology-based metrics. It enables:

- Evaluating how completely a survey captures routing behavior.
- Comparing survey systems on behavioral coverage.
- Guiding design of forwarding-aware probing algorithms.

Future Work

- Extend to **IPv6** using the IPv6 Hitlist as reference.
- Explore vantage point overlap and metric bias.
- Use the Forwarding Score to improve survey algorithms.

Acknowledgments

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